PROJECT EXPO 2026 – Engineer's Day Celebration

Objective of this EXPO:

- Foster innovation by solving real-world engineering problems using Electronics, Embedded Systems, IoT, AI/ML, PCB Design, Python, and Communication technologies.
- Encourage industry-standard product development through design, prototyping, testing, documentation, and timely project execution.
- Provide opportunities to interact with industry experts, alumni, and faculty mentors to gain practical knowledge and professional guidance.
- Develop technical excellence, teamwork, leadership, project management, and professional communication skills to prepare students for future engineering careers.

NOTE: “Projects may be based on Hardware, Software, Embedded Systems, IoT, VLSI, AI, or other interdisciplinary innovations”

DO's

1. Build Original Solutions

- Develop an original solution that addresses the given problem statement.
- Properly acknowledge all third-party libraries, datasets, and references used.

2. Follow Engineering Practices

- Maintain a structured design process including requirement analysis, architecture, implementation, testing, and validation.
- Follow good coding standards, hardware safety practices, and proper documentation.

3. Maintain Version History

- Use Git/GitHub or another version control system throughout development.
- Keep project milestones and progress records.

4. Meet Deadlines

- Submit all deliverables before the specified deadline.
- Late submissions may attract penalties or disqualification.

5. Document Everything

Maintain proper documentation including:

- Problem Definition
- Literature Survey
- System Architecture
- Circuit Diagrams
- PCB Design (if applicable)

- Flowcharts
- Algorithms
- Testing Results
- Bill of Materials
- Future Scope
- Etc..

6. Demonstrate Working Prototype

- The final demonstration should showcase a functional prototype.
- Teams should be able to explain every subsystem of their project.

7. Professional Communication

- Every team member should actively participate during reviews.
- Present ideas clearly using technical terminology and engineering justification.

8. Seek Mentorship

- Interact with faculty, alumni, and industry experts.
- Incorporate constructive feedback into subsequent iterations.

DON'Ts

1. No Plagiarism

- Do not copy projects from GitHub, YouTube, previous competitions, or commercial products.
- Any plagiarism may lead to immediate disqualification.

2. Do Not Purchase Ready-Made Projects

- Purchased or pre-built academic projects are strictly prohibited.

3. Avoid Last-Minute Development

- Projects developed only before the final evaluation without documented progress may receive lower scores.

4. Do Not Manipulate Results

- Fabricated test data, fake demonstrations, or edited videos without actual implementation are prohibited.

5. Do Not Ignore Documentation

- Poor or incomplete documentation may affect evaluation even if the prototype functions correctly.

6. Avoid Unsafe Practices

- Exposed wiring, overloaded power supplies, unsafe batteries, or hazardous demonstrations are not allowed.

7. Do Not Miss Mandatory Reviews

- Teams failing to attend scheduled design reviews or progress evaluations without prior approval may be disqualified.

Project Submission Timeline and List of Deliverables (Tentative):

Phase	Duration	Date	Deliverables
Registration	-	**Until 24 July**	Team Registration
Phase 1	Week 1	**25 Jul – 31 Jul**	Problem Identification & Research
Phase 2	Week 2	**01 Aug – 07 Aug**	Concept Development & System Design
Phase 3	Week 3-4	**08 Aug – 21 Aug**	Hardware/Software Design
Phase 4	Week 5-6	**22 Aug – 04 Sep**	Prototype Development
Phase 5	Week 7	**05 Sep – 10 Sep**	Testing & Validation
Final Phase	Week 8	**11 Sep – 14 Sep**	Final Submission & Presentation Preparation
Grand Finale	Show Day	**15 September (Engineers' Day)**	Live Demonstration & Evaluation

Phase - 1, Deliverables

- Problem Statement
- Abstract (200–300 words)
- Objectives
- Scope of the Project
- Expected Outcomes
- Literature Survey
- Existing Solutions Analysis
- Gap Analysis
- References
- Git Installation - configuration, Mastering GITHUB (Students, Faculties and Mentors)

Submit

- Technical Report (PDF), Editable DOCX, References, Literature Review Table, Problem Statement

Maximum Pages

10–15 Pages

Phase 2 – Ideation & Concept Development

Deliverables

- Brainstorming Report
- Idea Prioritization Matrix
- Empathy Map
- User Persona (if applicable)
- Proposed Solution
- Functional Requirements
- Non-functional Requirements
- Project Planning (Gantt Chart or Timeline)
- Risk Analysis

Submit

- Brainstorming Report, Empathy Map, Functional Requirements, Risk Matrix, Gantt Chart

Supported Formats

- PDF, Draw.io, PNG

Phase 3 – System Design Review, Deliverables

System Architecture

- Overall Solution Architecture
- Data Flow Diagram
- Functional Block Diagram
- Hardware-Software Partitioning

Hardware Design

- Final Block Diagram
- Circuit Schematics
- Component Selection
- Bill of Materials (BOM)
- PCB Layout (if applicable)

Software Design

- Flowcharts
- Algorithms
- Communication Protocols
- Database Structure (if applicable)

Simulation

- MATLAB / LTspice / Proteus / Tinkercad / Wokwi / KiCad Simulation
- Initial Results

Phase 4 – Prototype Development Review (PDR),**Deliverables**

- Working Prototype (Version 1)
- Embedded Software
- PCB Fabrication Status (if applicable)
- Mechanical Assembly (if applicable)
- Hardware Integration
- Software Integration
- GitHub Repository Link
- Weekly Progress Report

Submit

- GitHub Repository
- Prototype Images
- Prototype Video
- Source Code
- Weekly Progress Report

Video Duration: 3–5 Minutes

Phase 5 – Testing & Validation Review (TVR)

Deliverables

- Functional Testing Report
- Performance Evaluation
- Test Cases
- Error Analysis
- Debugging Report
- Power Consumption Analysis
- Reliability Testing
- Future Improvements

Final Submission

NOTE: VIVA is Included, Conducted in online by sept 13 or 14 (Tentative)

Technical Documents

- Final Project Report, Project Documentation, User Manual, Installation Guide,Maintenance Guide

Design Files

- Source Code, GitHub Repository, Circuit Schematics, PCB Design Files, CAD Files (if applicable), Simulation Files

Presentation Materials

- Project PPT, Demonstration and Explanation Video (3–5 min), "**Live Prototype**"

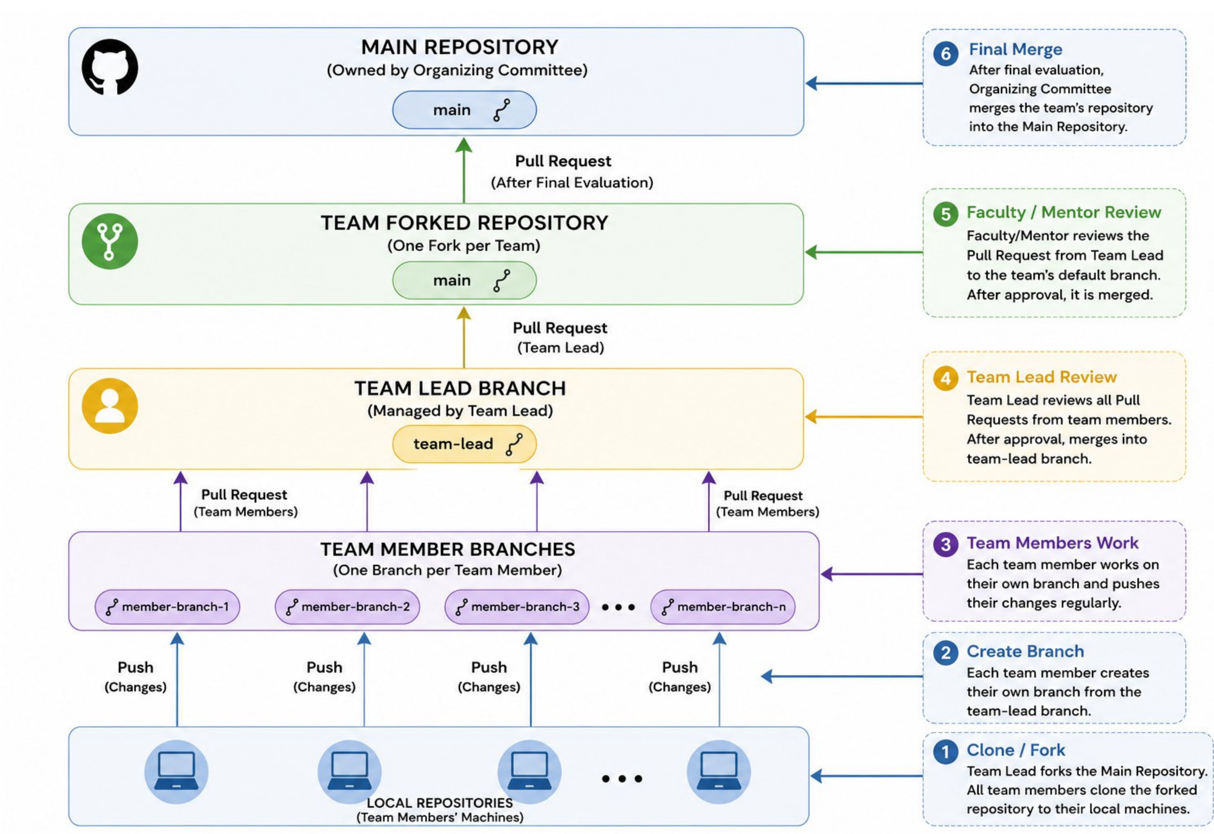
Business & Innovation

- Market Analysis, Cost Estimation, Business Model, Commercialization Strategy, Market Pitch, Future Scope, Patent / Research Scope (if applicable)

GIT FLOW:

- **Main Repository** (owned by Organizing Committee)
- **One Fork per Team**
- **One Branch per Team Member**
- **Team Lead reviews team members' Pull Requests**
- **Faculty/Mentor reviews Team Lead's Pull Request to the team's default branch**
- **After final evaluation, Organizing Committee merges the team's repository into the Main Repository**

NOTE: You can get help from Faculties or Mentors, Recommended: Learn from youtube and do yourself



Purpose	Recommended Tool
Documentation	Microsoft Word / LibreOffice Writer / Google Docs
Presentations	PowerPoint / Google Slides / LibreOffice Impress
Diagrams	Draw.io (diagrams.net)
Brainstorming	FigJam Free / Excalidraw
Gantt Chart	GanttProject / TeamGantt Free

Version Control	Git + GitHub
UI Design	Figma (Free)
Circuit Simulation	LTspice
Digital Electronics	Logisim Evolution
PCB Design	KiCad
Embedded Simulation	Wokwi/Proteus
Arduino Development	Arduino IDE
STM32 Development	STM32CubeIDE
Raspberry Pi	Raspberry Pi OS + VS Code
Python	VS Code / PyCharm Community
AI/ML	Google Colab
Data Analysis	Jupyter Notebook
Database	SQLite / MySQL Community
Flowcharts	Draw.io
UML Diagrams	PlantUML / Draw.io

NOTE: Each Team must have a faculty and Mentor